

Christopher Tastad

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TECHNICAL SUMMARY

Senior-level bioinformatics engineer with 5+ years architecting enterprise-scale genomic platforms and establishing computational biology standards. Leader in AWS infrastructure design (0.5PB+ datasets), single-cell and spatial transcriptomics methodology development, and containerized workflow systems. Proven track record of technical innovation including industry partnerships with 10x Genomics, pioneering spatial analysis pipelines, and creating foundational frameworks adopted across international research consortia.

TECHNICAL EXPERIENCE

Lead Data Engineer

Feb 2021 – Present

NIDDK IBD Genetics Consortium

New York, NY

- Architect and operate end-to-end enterprise-scale AWS infrastructure across 0.5PB+ genomic datasets including EC2 compute clusters, S3/Glacier storage systems, RDS databases, ELB load balancing, VPC networking, WAF security, EFS file systems, ECR, IAM, CloudFront, and SES supporting multiple public domains (ibdgc.org, ibdgenetics.org).
- Engineered comprehensive ID harmonization platform (IDhub) unifying legacy sample/subject IDs across 20 years of consortium recruitment, developing PostgreSQL database, REST API, global subject ID resolution service, and multi-staged curation workflows with audit logging.
- Implemented enterprise Sourcegraph deployment and served as admin for secure team LLM access, pioneering integration of AI-assisted development tools in bioinformatics workflows and establishing institutional standards for responsible AI adoption in genomic analysis.
- Deployed ultra-fast, web-based genomic annotation platform in AWS for WES/WGS analysis, reducing annotation time from days to hours for consortium-wide sequencing projects, making detailed genomic insights fast and widely accessible.
- Implemented FAIR-compliant data distribution architecture using Open Science Framework, fundamentally improving accessibility and meeting NIH data sharing requirements for 0.5PB+ genomic datasets across international research community.
- Established cloud-hosted automated subject identification system serving as central source of truth for 20-year longitudinal study recruitment and data collection, creating systematic data governance where none existed.
- Implemented systematic deployment standards using GitHub Actions for code and data process deployments, establishing reliability frameworks and reducing technical debt across international collaborative infrastructure.
- Implemented comprehensive data governance architecture focused on isolating PHI data within respective compute and storage environments, preventing contamination and ensuring compliance across institutional boundaries.
- Developed automated metadata management systems using LabKey sample management API, reducing manual curation overhead while establishing systematic data stewardship practices.

Senior Manager, Bioinformatics

Oct 2023 – Present

Icahn School of Medicine at Mount Sinai, Cho Lab

New York, NY

- Serve as tech lead on all methodological innovation, establishing leading analytical frameworks across dozen+ major project initiatives including evolving single-cell processing, automated cell typing, multi-omic data, spatial transcriptomics, and emerging genomic platforms. Create new computational biology standards through extensive testing, validation, pipeline deployment, and custom tool development.
- Pioneer spatial transcriptomics methodology extending Sopa for highly scalable execution and developing Parallax pipeline for noise correction, context-aware spatially differential expression, and downstream spatial analysis. Establish standard for foundational analytical frameworks enabling first-of-its-kind publications in IBD research.
- Developed official Mount Sinai institutional execution profile for nf-core pipelines, creating standardized HPC configuration endorsed by core scientific computing team and accepted as open-source contribution to nf-core community codebase.
- Led institution-first Xenium spatial analyzer device deployment including custom high-power supply configuration, full security review, automated data product distribution, and system networking implementation, establishing institutional deployment standards for emerging spatial platforms.
- Implement comprehensive data lifecycle management including storage class transition policies, 3-2-1 backup paradigm, data version control (DVC), and project-specific data registries. Achieve zero data loss incidents over 4+ years while dramatically reducing technical debt across 0.5PB+ managed data.
- Selected as 10x Genomics beta partner for cloud hosting feature development in Xenium Explorer analysis application, directly influencing product roadmap through technical feedback and validation of cloud-specific hosting capabilities in emerging spatial genomics platforms.

Bioinformatician II

Feb 2021 – Oct 2023

Icahn School of Medicine at Mount Sinai, Cho Lab

New York, NY

- Established standardized single-cell preprocessing paradigm (CoreSC) creating unified statistical methodology across 15+ projects where none previously existed, fundamentally improving reproducibility and eliminating analysis methods inconsistencies.

- Constructed comprehensive scientific analysis application stack including containerized workflows, automated data distribution, and several multi-user self-hosted services, fundamentally lowering barriers to access and accelerating discovery outcomes for non-technical biologists.
- Pioneered internal containerization paradigm using Docker/Kubernetes for application deployment, enabling non-technical users to deploy reproducible analysis environments in high performance compute resources.
- Sourced, built, and deployed purpose-built \$60k+ on-prem supplementary compute infrastructure including custom nodes, expanding team computational capacity 3x at fraction of commercial alternatives.
- Established strategic resource accessibility initiative deploying Digital Slide Archive, cloud storage integration, and containerization frameworks specifically to overcome substantial resource barriers introduced by growth of imaging and spatial transcriptomics workflows.

Graduate Research Assistant

May 2019 – Feb 2021

University of Minnesota Dept. of OBGYN, Starr Lab

Minneapolis, MN

- Executed full rewrite of cancer gene web app: deployed new ETL framework, reduced codebase, removed need for SQL database.
- Applied bioinformatic focus into copy number variation on largest of its kind single-cell sequencing human cancer dataset.
- Developed pipeline to automate analysis process from raw data to figures, incorporating HMM and FastCall algorithms.

Graduate Research Assistant

Jan 2016 – Aug 2017

National Marrow Donor Program, Bioinformatics

Minneapolis, MN

- Employed scientific method as an investigator in a corporate R&D department with a focus in hematological cancers.
- Took both basic and clinical approach to investigation in cell therapy, immunology, and organ transplant genetics.
- Aided development of initiative around IP generation using existing company technology and 13 million patient database.

Research Associate

Dec 2007 – Jan 2013

The Stem Cell Institute, Asakura Lab

Minneapolis, MN

- Performed basic science research with a focus in skeletal muscle development, receiving 4 co-authorships.
- Well versed in full spectrum of molecular biology, histology and cell biology wet lab techniques and diagnostics.
- Authored significant writing efforts for group manuscripts and major grant awards, including a successful R01 bid.

TECHNICAL ACHIEVEMENTS

Architected enterprise AWS infrastructure supporting 0.5PB+ genomic datasets across international consortium
 Developed novel spatial transcriptomics methodology (Parallax pipeline) enabling first-of-its-kind IBD research publications
 Created standardized single-cell preprocessing framework (CoreSC) adopted across 15+ research projects
 Engineered comprehensive ID harmonization platform (IDhub) unifying 20 years of consortium data
 Pioneered enterprise AI integration deployment and secure LLM access for bioinformatics team workflows
 Selected as 10x Genomics beta partner for emerging spatial genomics platform development
 Achieved zero data loss incidents over 4+ years while managing 0.5PB+ research datasets
 Pioneered containerization paradigm enabling reproducible analysis environments for non-technical users
 Developed official institutional nf-core nextflow execution profile accepted as open-source contribution

TECHNICAL EXPERTISE

Cloud & Infrastructure Architecture: AWS (EC2, S3, Glacier, RDS, ELB, VPC, WAF, EFS, ECR, IAM, CloudFront, SES, Route53), Docker, Kubernetes, Nginx, Apache, HPC (SLURM, PBS, LSF), Linux (Ubuntu, RHEL, CentOS)
Programming & Development: Python, R, BASH/Shell, SQL, LaTeX, HTML, CSS, JavaScript, XML, JSON, YAML, Git/GitHub, CI/CD (GitHub Actions)
Data Management & Platforms: PostgreSQL, MySQL, Oracle, Data Version Control (DVC), Globus, REDCap, LabKey, OSF, Dataverse, ETL frameworks
Genomics & Bioinformatics Platforms: Nextflow, Snakemake, 10x Genomics (Cellranger, Space Ranger), Seurat, Scanpy, CellTypist, samtools, bwa-mem, bedtools, VarScan2, GATK, Bystro, nf-core
Spatial Transcriptomics & Advanced Methods: Visium/Xenium analysis, Sopa framework, Parallax pipeline development, Digital Slide Archive, Multi-omic integration, CITE-seq, ATAC-seq, CRISPRi
Analysis & Statistical Computing: R (ggplot2, dplyr, Seurat, rmarkdown, shiny), Python (pandas, numpy, scipy, matplotlib, seaborn, scanpy), Statistical modeling, Machine learning, PCA, UMAP, t-SNE
Platform & Service Development: CellxGene, Digital Slide Archive, JupyterHub, Sourcegraph, Ollama, WordPress, REST API development, Containerization
Data Types & Methodologies: Single-cell RNA-seq, Spatial transcriptomics, ATAC-seq, CITE-seq, CRISPRi, Nanopore, WGS/WES, GWAS, AIRR-seq, Bulk RNA-seq, Copy number variation analysis

EDUCATION

University of Minnesota

Minneapolis, MN

Master of Science in Bioinformatics and Computational Biology

2021

University of Minnesota

Minneapolis, MN

Bachelor of Science in Biochemistry, College of Biological Sciences

2012